Elham Aghakhani

Philadelphia, PA

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Personal Website Linkedin

Google Scholar Github

Education

PhD **Drexel University**, Information Science — Jan 2024 - Expected June 2028

MSc **K.N Toosi University of Technology**, Computer Science, AI — Oct 2020 - Sep 2023

- **Coursework:** Machine Learning, Pattern Recognition, Big Data Analysis, Reinforcement Learning
- **Thesis:** Automatic Persian Text Summarization using Pre-trained Language Models and Reinforcement Learning.

BSc **K. N. Toosi University of Technology**, Computer Science — Sep 2015 - Sep 2019

- **Thesis:** Implementing an Automatic System to Collect News from News Agencies.

Research Interests

- Evaluation LLMs in real-world settings
- Human-centered AI
- Computational Social Science and Sociotechnical Systems
- Behavioral modeling and observational studies

Publications

- **Aghakhani E**, Wang L, Washington KT, Demiris G, Huh-Yoo J, Rezapour R. (2025). From Conversation to Automation: Leveraging LLMs for Problem-Solving Therapy Analysis. Findings of the Association for Computational Linguistics (**ACL**), Vienna, Austria, pp. 25189–25207.
- Bouzoubaa, L., **Aghakhani, E.**, Rezapour, R. S. (2025). Phenotypes of stigma expressed by people who use drugs on Reddit. **Social science & medicine**, 118889.
- Namvarpour M*, **Aghakhani E***, Ekstrand MD, Rezapour R, Razi A. (2025). *The Evolving Landscape of Youth Online Safety: Insights from News Media Analysis*. **WebSci'25**. *Equal contribution.
- Wang L, Lynch D, **Aghakhani E**, Demiris G, Washington K, Rezapour R, Huh-Yoo J. (2025) *AnnoLoom: Augmenting Codebook Generation and Annotation with Large Language Models.* Proceedings of the Association for Information Science and Technology.

- Bouzoubaa L, **Aghakhani E**, Rezapour R. (2024). Words Matter: Reducing Stigma in Online Conversations about Substance Use with Large Language Models. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (**EMNLP**) Nov 2024.
- Bouzoubaa L, **Aghakhani E**, Song M, Trinh M, Rezapour R. (2024). Decoding the Narratives: Analyzing Personal Drug Experiences Shared on Reddit. The 62nd Annual Meeting of the Association for Computational Linguistics (**ACL**) Findings.

Experience

Instructor of Record

- **Drexel University**: Introduction to Data Science (INFO103-900) – Summer 2025

Teaching Assistant

- **Drexel University**:
  - Web Systems and Services (INFO151) - Fall 2025
  - Social Collaborative and Computing (INFO616) – Winter 2025
- **K. N. Toosi University of Technology**:
  - Probabilistic Graphical Models – 2021–2022
  - Machine Learning – 2021–2022
  - Statistical Pattern Recognition – 2021–2022

Research Assistant

- **Drexel University** (Jan 2024 – Present)
  - Designed expert-informed annotation rubrics for mental health–related language
  - Designed and analyzed large-scale observational studies using social media and news data.
  - Conducted human-centered analyses of online behavior, stigma, and AI-assisted interventions.
  - Designed prompt-based and simulation-style experiments using LLM agents.
  - Evaluated generative AI systems for social and behavioral outcomes.
  - Used Python for data cleaning, statistical analysis, and modeling.

Industry Experience

- **ERON Company**, Web Developer (Sep 2019 – Mar 2020)
  - Designed and built databases, web applications, and webpages using backend technologies.
- **VADA House of Mobile**, Web Developer (Jan 2019 – Sep 2019)
  - Designed and implemented web applications and APIs using PHP and Laravel.

Peer Review Activities

- Reviewer for: *ACL 2026*, *WebSci'2026*, *ICWSM 2025*, *ACL 2025*, *NAACL 2024*, *EMNLP 2024*

Skills

- **Data Analysis**: Python, Pandas, NumPy, scikit-learn, PyTorch
- **Generative AI and NLP**: HuggingFace Transformers, LLM fine-tuning, prompt engineering
- **Web and Data Collection**: Web scraping, APIs, SQL, PHP, Laravel, basic JavaScript
- **Research Tools**: Git, GitHub, LaTeX

Certifications

Natural Language Processing, Coursera (Aug 2021):
- Credential ID: 5WUJGRXXP22V

Machine learning in python with scikit-learn, France Universite Numerique (Jul 2021):
- Credential ID: 4TD52DHM5TCD

Deep Learning workshop, IEEE Iran section (Mar 2021):
- Credential ID: 147474

Awards, Fellowships, and Highlights

Selected to attend the 8th Annual Summer Institute in Computational Social Science, University of Pennsylvania — Philadelphia, PA, July 2024